

# Universidad Tecnológica de Emiliano Zapata (UTEZ)

Carrera: **Ingeniería en desarrollo y gestión de software**

Materia: **Extracción de conocimiento de Bases de Datos**

Alumno: **Ruiz Santos Orlando**

Profesor: **Eliel Rodriguez**

*Examen U2*

Grado: **9º**

Grupo: **A**

Fecha: **05/07/26**

## 1. Captura del repositorio con la estructura solicitada:

The screenshot shows a web browser window displaying a GitHub repository. The browser's address bar shows the URL `github.com/RuizOr/Unidad2_ETL_Biblioteca_RuizSantosOrlando`. The repository name is `Unidad2_ETL_Biblioteca_RuizSantosOrlando` and it is marked as `Public`. The repository has 0 forks, 0 stars, and 0 watchers. The main branch is `main`, and there is 1 branch and 0 tags. The repository contains a commit by `RuizOr` with the message `feat: tarea terminada`, dated 3 minutes ago. The commit includes a table of files and folders:

| File/Folder             | Commit Message                             | Time Ago      |
|-------------------------|--------------------------------------------|---------------|
| <code>data</code>       | <code>feat: estructura del proyecto</code> | 1 hour ago    |
| <code>evidencias</code> | <code>feat: tarea terminada</code>         | 3 minutes ago |
| <code>scripts</code>    | <code>feat: tarea terminada</code>         | 3 minutes ago |
| <code>sql</code>        | <code>feat: tarea terminada</code>         | 3 minutes ago |
| <code>README.md</code>  | <code>feat: tarea terminada</code>         | 3 minutes ago |

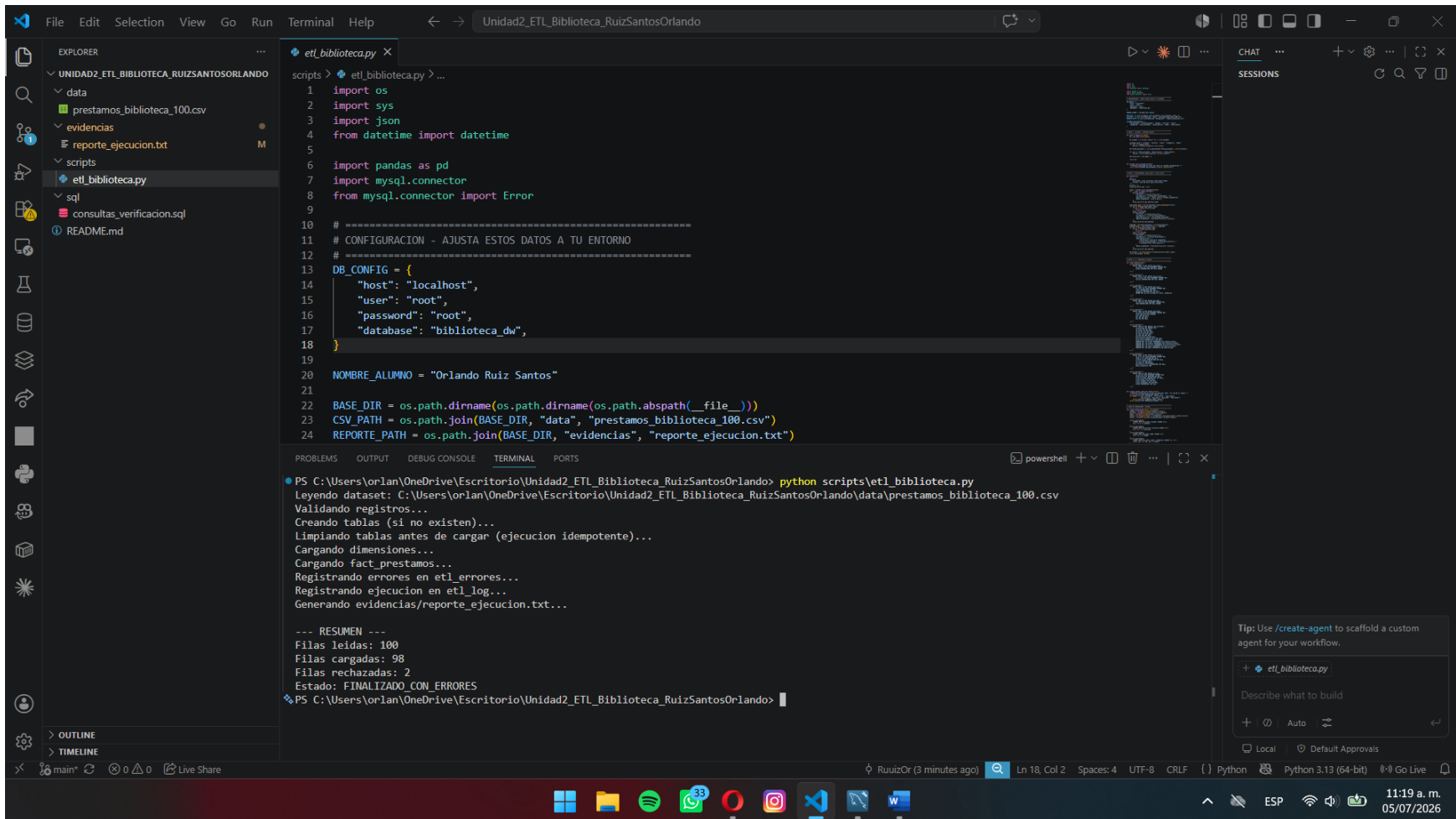
The repository also has a `README` file. The `README` content is as follows:

### Unidad 2 - ETL Prestamos de Biblioteca

#### 1. Objetivo del proyecto

The right sidebar of the repository page shows the `About` section, which states: `No description, website, or topics provided.` Below this, there are links for `Readme`, `Activity`, `0 stars`, `0 watching`, and `0 forks`. The `Releases` section states: `No releases published` and provides a link to `Create a new release`. The `Packages` section states: `No packages published` and provides a link to `Publish your first package`. The `Contributors` section shows 1 contributor.

## 2. Captura de ejecución del script:



The screenshot displays the Visual Studio Code interface with a Python script named `etl_biblioteca.py` open in the editor. The script is located in the `scripts` directory of a project named `UNIDAD2_ETL_BIBLIOTECA_RUIZSANTOSORLANDO`. The script's content is as follows:

```
1 import os
2 import sys
3 import json
4 from datetime import datetime
5
6 import pandas as pd
7 import mysql.connector
8 from mysql.connector import Error
9
10 # =====
11 # CONFIGURACION - AJUSTA ESTOS DATOS A TU ENTORNO
12 # =====
13 DB_CONFIG = {
14     "host": "localhost",
15     "user": "root",
16     "password": "root",
17     "database": "biblioteca_dw",
18 }
19
20 NOMBRE_ALUMNO = "Orlando Ruiz Santos"
21
22 BASE_DIR = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
23 CSV_PATH = os.path.join(BASE_DIR, "data", "prestamos_biblioteca_100.csv")
24 REPORTE_PATH = os.path.join(BASE_DIR, "evidencias", "reporte_ejecucion.txt")
```

The terminal window shows the execution of the script using the command `python scripts\etl_biblioteca.py`. The output indicates that the dataset `C:\Users\orlan\OneDrive\Escritorio\Unidad2_ETL_Biblioteca_RuizSantosOrlando\data\prestamos_biblioteca_100.csv` was read, records were validated, tables were created (if they didn't exist), tables were cleaned (for idempotent execution), dimensions were loaded, `fact_prestamos` was loaded, errors were registered in `etl_errores`, execution was logged in `etl_log`, and evidence was generated in `reporte_ejecucion.txt`. A summary of the execution is provided below:

```
--- RESUMEN ---
Filas leídas: 100
Filas cargadas: 98
Filas rechazadas: 2
Estado: FINALIZADO CON ERRORES
```

The status bar at the bottom of the editor shows the file encoding as UTF-8, the line and column position as Ln 18, Col 2, and the active interpreter as Python 3.13 (64-bit).

### 3. Captura de las tablas creadas en MySQL:

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' list, with 'biblioteca\_dw' selected. The main editor window shows the SQL command 'SHOW TABLES;'. The bottom panel displays the 'Output' window with the following results:

| #  | Time     | Action                                                                                                        | Message            | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------------------------------------------|--------------------|-----------------------|
| 15 | 11:09:50 | SELECT s.sede, AVG(f.dias_prestamo) AS promedio_dias_prestamo FROM fact_prestamos f JOIN dim_se...            | 4 row(s) returned  | 0.000 sec / 0.000 sec |
| 16 | 11:10:30 | SELECT l.libro, l.categoria, SUM(f.total_multa) AS total_multa FROM fact_prestamos f JOIN dim_libro l ON f... | 5 row(s) returned  | 0.000 sec / 0.000 sec |
| 17 | 11:11:47 | SELECT f.id_prestamo, fe.fecha, a.alumno, c.carrera, l.libro, l.categoria, s.sede, f.total...                 | 98 row(s) returned | 0.000 sec / 0.000 sec |
| 18 | 11:12:03 | SELECT s.sede, COUNT(*) AS total_prestamos FROM fact_prestamos f JOIN dim_sede s ON f.id_sede = s...          | 4 row(s) returned  | 0.015 sec / 0.000 sec |
| 19 | 11:20:38 | SHOW TABLES                                                                                                   | 8 row(s) returned  | 0.015 sec / 0.000 sec |

## 4. Captura de `fact\_prestamos` con datos:

MySQL Workbench

Local instance MySQL8...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- alumnos\_db
- biblioteca
- biblioteca\_dw**
  - Tables
  - Views
  - Stored Procedures
  - Functions
- chat
- empleados
- employee\_db
- empresa
- hospiorlando
- hospitalsaludvital
- linkedin\_data
- mascotasa
- mi\_sistema\_tutores
- perros
- productos\_db
- rac\_db
- ris
- sakila
- sftv
- sys
- tienda
- tricks
- vehiculos

Administration Schemas

Information

Schema: biblioteca\_dw

SQL File 13\* pacientes tratamientos tratamientos SQL File 6\* SQL File 7\* SQL File 8\*

Limit to 1000 rows

1 \* `SELECT * FROM fact_prestamos;`

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

|   | id_prestamo | id_fecha | id_alumno | id_carrera | id_libro | id_sede | dias_prestamo | multa_diaria | total_multa |
|---|-------------|----------|-----------|------------|----------|---------|---------------|--------------|-------------|
| ▶ | 5001        | 1        | 2         | 2          | 1        | 1       | 3             | 0.00         | 0.00        |
|   | 5002        | 2        | 10        | 4          | 2        | 3       | 5             | 5.00         | 25.00       |
|   | 5003        | 3        | 19        | 3          | 3        | 4       | 7             | 10.00        | 70.00       |
|   | 5004        | 4        | 16        | 5          | 4        | 2       | 2             | 15.00        | 30.00       |
|   | 5005        | 5        | 9         | 1          | 5        | 1       | 4             | 0.00         | 0.00        |
|   | 5006        | 6        | 3         | 2          | 6        | 3       | 6             | 5.00         | 30.00       |
|   | 5007        | 7        | 11        | 4          | 7        | 4       | 8             | 10.00        | 80.00       |
|   | 5008        | 8        | 8         | 3          | 8        | 2       | 1             | 15.00        | 15.00       |
|   | 5009        | 9        | 4         | 5          | 9        | 1       | 9             | 0.00         | 0.00        |
|   | 5010        | 10       | 15        | 1          | 10       | 3       | 10            | 5.00         | 50.00       |
|   | 5011        | 11       | 18        | 2          | 1        | 4       | 3             | 10.00        | 30.00       |
|   | 5012        | 12       | 1         | 4          | 2        | 2       | 5             | 15.00        | 75.00       |
|   | 5013        | 13       | 6         | 3          | 3        | 1       | 7             | 0.00         | 0.00        |
|   | 5014        | 14       | 12        | 5          | 4        | 3       | 2             | 5.00         | 10.00       |
|   | 5015        | 15       | 5         | 1          | 5        | 4       | 4             | 10.00        | 40.00       |
|   | 5016        | 16       | 17        | 2          | 6        | 2       | 6             | 15.00        | 90.00       |

fact\_prestamos 9 x

Output

Action Output

| #  | Time     | Action                                                                                                        | Message            | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------------------------------------------|--------------------|-----------------------|
| 16 | 11:10:30 | SELECT l.libro, l.categoria, SUM(f.total_multa) AS total_multa FROM fact_prestamos f JOIN dim_libro l ON f... | 5 row(s) returned  | 0.000 sec / 0.000 sec |
| 17 | 11:11:47 | SELECT f.id_prestamo, fe.fecha, a.alumno, c.carrera, l.libro, l.categoria, s.sede, f.total...                 | 98 row(s) returned | 0.000 sec / 0.000 sec |
| 18 | 11:12:03 | SELECT s.sede, COUNT(*) AS total_prestamos FROM fact_prestamos f JOIN dim_sede s ON f.id_sede = s...          | 4 row(s) returned  | 0.015 sec / 0.000 sec |
| 19 | 11:20:38 | SHOW TABLES                                                                                                   | 8 row(s) returned  | 0.015 sec / 0.000 sec |
| 20 | 11:22:01 | SELECT * FROM fact_prestamos LIMIT 0, 1000                                                                    | 98 row(s) returned | 0.000 sec / 0.000 sec |

Object Info Session

11:22 a. m.  
05/07/2026

## 5. Captura de `etl\_errores` con los 2 errores:

MySQL Workbench

Local instance MySQL8... x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- alumnos\_db
- biblioteca
- biblioteca\_dw**
  - Tables
  - Views
  - Stored Procedures
  - Functions
- chat
- empleados
- employee\_db
- empresa
- hospiorlando
- hospitalaludvital
- linkedin\_data
- mascotasa
- mi\_sistema\_tutores
- perros
- productos\_db
- rac\_db
- ris
- sakila
- sfti
- sys
- tienda
- tricks
- vehiculos

Administration Schemas

Information

Schema: biblioteca\_dw

Object Info Session

SQL File 13\* pacientes tratamientos tratamientos SQL File 6\* SQL File 7\* SQL File 8\* x

Limit to 1000 rows

```
1 SELECT * FROM etl_errores;
```

Result Grid

| # | id_error | fecha_error         | archivo_origen               | fila_csv | id_registro | descripcion_error                      | datos_originales                       |
|---|----------|---------------------|------------------------------|----------|-------------|----------------------------------------|----------------------------------------|
| 1 | 1        | 2026-07-05 11:19:38 | prestamos_biblioteca_100.csv | 101      | 5002        | id_prestamo duplicado                  | {"id_prestamo": 5002, "fecha_presta... |
| 2 | 2        | 2026-07-05 11:19:38 | prestamos_biblioteca_100.csv | 100      | 5099        | total_multa incorrecto (esperado=70... | {"id_prestamo": 5099, "fecha_presta... |

etl\_errores 10 x

Output

Action Output

| #  | Time     | Action                                                                                               | Message            | Duration / Fetch      |
|----|----------|------------------------------------------------------------------------------------------------------|--------------------|-----------------------|
| 17 | 11:11:47 | SELECT f.id_prestamo, fe.fecha, a.alumno, c.carrera, l.libro, l.categoria, s.sede, f.total...        | 98 row(s) returned | 0.000 sec / 0.000 sec |
| 18 | 11:12:03 | SELECT s.sede, COUNT(*) AS total_prestamos FROM fact_prestamos f JOIN dim_sede s ON f.id_sede = s... | 4 row(s) returned  | 0.015 sec / 0.000 sec |
| 19 | 11:20:38 | SHOW TABLES                                                                                          | 8 row(s) returned  | 0.015 sec / 0.000 sec |
| 20 | 11:22:01 | SELECT * FROM fact_prestamos LIMIT 0, 1000                                                           | 98 row(s) returned | 0.000 sec / 0.000 sec |
| 21 | 11:22:49 | SELECT * FROM etl_errores LIMIT 0, 1000                                                              | 2 row(s) returned  | 0.000 sec / 0.000 sec |

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Apply Revert Context Help Snippets

11:22 a. m.  
05/07/2026

## 6. Captura de `etl\_log`:

MySQL Workbench

Local instance MySQL8...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- alumnos\_db
- biblioteca
- biblioteca\_dw**
  - Tables
  - Views
  - Stored Procedures
  - Functions
- chat
- empleados
- employee\_db
- empresa
- hospiorlando
- hospitalaludvital
- linkedin\_data
- mascotas
- mi\_sistema\_tutores
- perros
- productos\_db
- rac\_db
- ris
- sakila
- sfti
- sys
- tienda
- tricks
- vehiculos

Administration Schemas

Information

Schema: biblioteca\_dw

Object Info Session

SQL File 13\* pacientes tratamientos tratamientos SQL File 6\* SQL File 7\* SQL File 8\*

Limit to 1000 rows

```
1 • SELECT * FROM etl_log;
```

Result Grid

|   | id_log | fecha_ejecucion     | archivo_origen               | filas_leidas | filas_cargadas | filas_rechazadas | estado                 |
|---|--------|---------------------|------------------------------|--------------|----------------|------------------|------------------------|
| ▶ | 1      | 2026-07-05 10:43:37 | prestamos_biblioteca_100.csv | 100          | 98             | 2                | FINALIZADO_CON_ERRORES |
|   | 2      | 2026-07-05 10:45:45 | prestamos_biblioteca_100.csv | 100          | 98             | 2                | FINALIZADO_CON_ERRORES |
|   | 3      | 2026-07-05 11:05:17 | prestamos_biblioteca_100.csv | 100          | 98             | 2                | FINALIZADO_CON_ERRORES |
|   | 4      | 2026-07-05 11:19:38 | prestamos_biblioteca_100.csv | 100          | 98             | 2                | FINALIZADO_CON_ERRORES |
| * |        |                     |                              |              |                |                  |                        |

etl\_log 11 x

Output

Action Output

| #  | Time     | Action                                                                                               | Message            | Duration / Fetch      |
|----|----------|------------------------------------------------------------------------------------------------------|--------------------|-----------------------|
| 18 | 11:12:03 | SELECT s.sede, COUNT(*) AS total_prestamos FROM fact_prestamos f JOIN dim_sede s ON f.id_sede = s... | 4 row(s) returned  | 0.015 sec / 0.000 sec |
| 19 | 11:20:38 | SHOW TABLES                                                                                          | 8 row(s) returned  | 0.015 sec / 0.000 sec |
| 20 | 11:22:01 | SELECT * FROM fact_prestamos LIMIT 0, 1000                                                           | 98 row(s) returned | 0.000 sec / 0.000 sec |
| 21 | 11:22:49 | SELECT * FROM etl_errores LIMIT 0, 1000                                                              | 2 row(s) returned  | 0.000 sec / 0.000 sec |
| 22 | 11:23:30 | SELECT * FROM etl_log LIMIT 0, 1000                                                                  | 4 row(s) returned  | 0.000 sec / 0.000 sec |

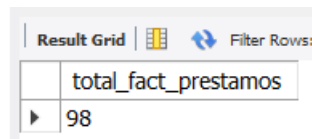
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Apply Revert Context Help Snippets

11:23 a. m.  
05/07/2026

## 7. Captura o evidencia de las consultas SQL:

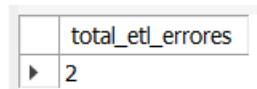
1. Cuantos registros hay en `fact\_prestamos`.



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The query 'total\_fact\_prestamos' is entered in the top bar. The result table has two columns: the first is empty, and the second contains the value '98'.

|   | total_fact_prestamos |
|---|----------------------|
| ▶ | 98                   |

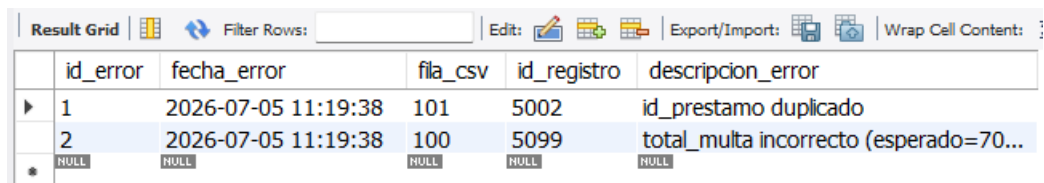
2. Cuantos registros hay en `etl\_errores`.



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The query 'total\_etl\_errores' is entered in the top bar. The result table has two columns: the first is empty, and the second contains the value '2'.

|   | total_etl_errores |
|---|-------------------|
| ▶ | 2                 |

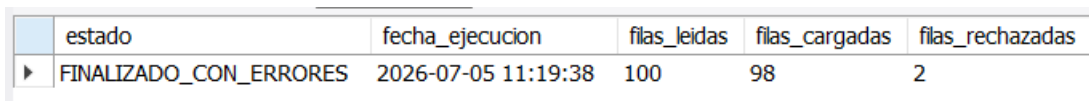
3. Que errores fueron registrados.



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button and various tool icons. The result table has six columns: id\_error, fecha\_error, fila\_csv, id\_registro, and descripcion\_error. There are two rows of data.

|   | id_error | fecha_error         | fila_csv | id_registro | descripcion_error                      |
|---|----------|---------------------|----------|-------------|----------------------------------------|
| ▶ | 1        | 2026-07-05 11:19:38 | 101      | 5002        | id_prestamo duplicado                  |
|   | 2        | 2026-07-05 11:19:38 | 100      | 5099        | total_multa incorrecto (esperado=70... |

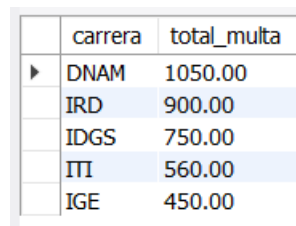
4. Cual fue el último estado registrado en `etl\_log`.



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The result table has five columns: estado, fecha\_ejecucion, filas\_leidas, filas\_cargadas, and filas\_rechazadas. There is one row of data.

|   | estado                 | fecha_ejecucion     | filas_leidas | filas_cargadas | filas_rechazadas |
|---|------------------------|---------------------|--------------|----------------|------------------|
| ▶ | FINALIZADO_CON_ERRORES | 2026-07-05 11:19:38 | 100          | 98             | 2                |

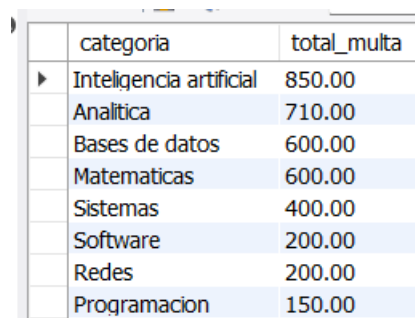
5. Total de multas por carrera.



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The result table has two columns: carrera and total\_multa. There are five rows of data.

|   | carrera | total_multa |
|---|---------|-------------|
| ▶ | DNAM    | 1050.00     |
|   | IRD     | 900.00      |
|   | IDGS    | 750.00      |
|   | ITI     | 560.00      |
|   | IGE     | 450.00      |

6. Total de multas por categoria de libro.



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The result table has two columns: categoria and total\_multa. There are eight rows of data.

|   | categoria               | total_multa |
|---|-------------------------|-------------|
| ▶ | Inteligencia artificial | 850.00      |
|   | Analitica               | 710.00      |
|   | Bases de datos          | 600.00      |
|   | Matematicas             | 600.00      |
|   | Sistemas                | 400.00      |
|   | Software                | 200.00      |
|   | Redes                   | 200.00      |
|   | Programacion            | 150.00      |



## 7. Promedio de días de préstamo por sede.

|   | sede                | promedio_dias_prestamo |
|---|---------------------|------------------------|
| ► | Biblioteca Central  | 6.2000                 |
|   | Sala Sur            | 6.0833                 |
|   | Sala Norte          | 4.8000                 |
|   | Laboratorio Digital | 4.5833                 |

## 8. Los 5 libros con mayor total de multa.

|   | libro                | categoria               | total_multa |
|---|----------------------|-------------------------|-------------|
| ► | Fundamentos de IA    | Inteligencia artificial | 850.00      |
|   | Estadística Aplicada | Matematicas             | 600.00      |
|   | Bases de Datos I     | Bases de datos          | 500.00      |
|   | Linux para Todos     | Sistemas                | 400.00      |
|   | Análisis de Datos    | Analítica               | 360.00      |

## 9. Prestamos detallados con fecha, alumno, carrera, libro, categoria, sede y total de multa.

| id_prestamo | fecha      | alumno         | carrera | libro                  | categoria               | sede                | total_multa |
|-------------|------------|----------------|---------|------------------------|-------------------------|---------------------|-------------|
| 5029        | 2026-06-01 | Daniel Torres  | ITI     | Análisis de Datos      | Analítica               | Biblioteca Central  | 0.00        |
| 5001        | 2026-06-01 | Ana Lopez      | IDGS    | Python Básico          | Programación            | Biblioteca Central  | 0.00        |
| 5057        | 2026-06-01 | Valeria Cruz   | IRD     | Linux para Todos       | Sistemas                | Biblioteca Central  | 0.00        |
| 5085        | 2026-06-01 | Laura Diaz     | DNAM    | Ingeniería de Software | Software                | Biblioteca Central  | 0.00        |
| 5086        | 2026-06-02 | Carlos Luna    | IDGS    | Estadística Aplicada   | Matematicas             | Sala Norte          | 30.00       |
| 5030        | 2026-06-02 | Paola Reyes    | DNAM    | Fundamentos de IA      | Inteligencia artificial | Sala Norte          | 50.00       |
| 5002        | 2026-06-02 | Luis Perez     | IRD     | Bases de Datos I       | Bases de datos          | Sala Norte          | 25.00       |
| 5058        | 2026-06-02 | Hector Medina  | IGE     | SQL Practico           | Bases de datos          | Sala Norte          | 5.00        |
| 5059        | 2026-06-03 | Natalia Flores | ITI     | Análisis de Datos      | Analítica               | Sala Sur            | 90.00       |
| 5003        | 2026-06-03 | Sofía Ruiz     | IGE     | Minería de Datos       | Analítica               | Sala Sur            | 70.00       |
| 5031        | 2026-06-03 | Roberto Silva  | IDGS    | Python Básico          | Programación            | Sala Sur            | 30.00       |
| 5087        | 2026-06-03 | Maria Solís    | IRD     | Linux para Todos       | Sistemas                | Sala Sur            | 80.00       |
| 5060        | 2026-06-04 | Oscar Herrera  | DNAM    | Fundamentos de IA      | Inteligencia artificial | Laboratorio Digital | 150.00      |
| 5088        | 2026-06-04 | Jorge Cano     | IGE     | SQL Practico           | Bases de datos          | Laboratorio Digital | 15.00       |
| 5004        | 2026-06-04 | Pedro Mata     | ITI     | Redes de Computad...   | Redes                   | Laboratorio Digital | 30.00       |
| 5032        | 2026-06-04 | Alejandra Mora | IRD     | Bases de Datos I       | Bases de datos          | Laboratorio Digital | 75.00       |
| 5033        | 2026-06-05 | Fernanda Rios  | IGE     | Minería de Datos       | Analítica               | Biblioteca Central  | 0.00        |
| 5005        | 2026-06-05 | Laura Diaz     | DNAM    | Ingeniería de Software | Software                | Biblioteca Central  | 0.00        |
| 5061        | 2026-06-05 | Ana Lopez      | IDGS    | Python Básico          | Programación            | Biblioteca Central  | 0.00        |
| 5089        | 2026-06-05 | Daniel Torres  | ITI     | Análisis de Datos      | Analítica               | Biblioteca Central  | 0.00        |

(hay más datos abajo, pero solo coloque captura de los primeros)

## 10. Conteo de préstamos por sede.

|   | sede                | total_prestamos |
|---|---------------------|-----------------|
| ► | Biblioteca Central  | 25              |
|   | Sala Norte          | 25              |
|   | Laboratorio Digital | 24              |
|   | Sala Sur            | 24              |

## 8. Breve explicación del flujo ETL:

El proceso comienza leyendo el archivo `prestamos_biblioteca_100.csv` con `pandas`, donde se estandarizan los nombres de columnas, se eliminan espacios en los campos de texto y se convierten `fecha_prestamo`, `dias_prestamo`, `multa_diaria` y `total_multa` a sus tipos correctos. Después se valida cada registro: se revisa que no existan nulos en columnas obligatorias, se detectan `id_prestamo` duplicados (conservando la primera aparición y rechazando la segunda) y se verifica que `total_multa`

sea igual a  $\text{dias\_prestamo} * \text{multa\_diaria}$ . Los registros que no pasan estas validaciones se guardan en `etl_errores` con la descripción del problema. Los 98 registros válidos se cargan a las tablas de dimensiones (`dim_alumno`, `dim_carrera`, `dim_libro`, `dim_sede`, `dim_fecha`) y a la tabla de hechos `fact_prestamos`. Antes de cada carga, el script limpia las tablas (excepto `etl_log`) para permitir ejecuciones repetidas sin duplicar información. Finalmente, se registra un resumen de la ejecución en `etl_log` (filas leídas, cargadas, rechazadas y estado) y se genera automáticamente el archivo `reporte_ejecucion.txt` como comprobante de que el proceso corrió realmente.